Exact Virtual Unknotting Barriers and a Classical–Virtual Complexity Gap

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Abstract

We introduce and compute the *virtual minimax barrier* $\mathrm{Top}_v(D)$ — the minimum crossing-number ceiling over all sequences of algebraic Reidemeister moves on Gauss words that unknot a diagram D — and contrast it with the *classical barrier* $\mathrm{Top}_c(D)$ restricted to planar-realizable moves. We present an exact algorithm, BoundedReidemeisterSearch, that computes $\mathrm{Top}_v(D)$ via bounded BFS with crossing-number pruning, and prove its correctness. Applying it to five hard unknot diagrams from the literature, we obtain $\mathrm{Top}_v(\text{Goeritz}) = 11$, $\mathrm{Top}_v(\text{Culprit}) = 10$, $\mathrm{Top}_v(D_{28}) = 28$, $\mathrm{Top}_v(D_{43}) = 43$, and $\mathrm{Top}_v(\text{Ochiai II}) = 45$. Our main result is a separation between virtual and classical unknotting complexity: for the D_{28} and D_{43} diagrams, $\mathrm{Top}_v(D) = \mathrm{cr}(D)$ but $\mathrm{Top}_c(D) = \mathrm{cr}(D)+3$ (by Burton et al. [4]), yielding a gap of 3 crossings in both cases. This provides rigorous examples from both known families of hard unknots (the Freedman–He–Wang generalization and the Goeritz generalization) where algebraic flexibility of Gauss words strictly reduces the minimax crossing barrier compared to geometric (planar) Reidemeister moves. We also show that D_{28} and Ochiai II admit *monotone-reducing* virtual unknotting paths using only R1-down and R2-down moves, reminiscent of Dynnikov's monotonic simplification for arc-presentations.

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1 Introduction

A *knot* is a piecewise-linear embedding of S^1 into the three-dimensional sphere $\mathbb{S}^3$, represented by a *diagram* D — a projection into $\mathbb{S}^2$ with transverse double points called *crossings*. The number of crossings is denoted $\mathrm{cr}(D)$. By Reidemeister's theorem [9], two diagrams represent equivalent knots if and only if one can be transformed into the other by a finite sequence of *Reidemeister moves* (R1, R2, R3) and their inverses.

The *Reidemeister graph* $R(K)$ of a knot K is the infinite graph whose vertices are $\mathbb{S}^2$-isotopy classes of diagrams of K, with edges connecting diagrams related by a single Reidemeister move. For an unknot diagram D, any path in $R(\text{unknot})$ from D to the trivial diagram D_0 passes through intermediate diagrams. Define

$$\mathrm{Top}(D) = \min_{\text{paths } D \to D_0} \max_{D' \in \text{path}} \mathrm{cr}(D')$$

to be the *minimax crossing number* of D. The *extra crossings* $m(D) = \mathrm{Top}(D) - \mathrm{cr}(D)$ measures the "hardness" of D as an unknot diagram [6, 1].

Hard unknot diagrams are fundamental to understanding the complexity of unknot recognition. Exponential upper bounds on $m(D)$ [3] were improved to quadratic by Dynnikov [2] and Lackenby [7]. On the lower bound side, Hass and Nowik exhibited infinite families requiring quadratically many moves but with $m = 0$ [4]. Burton et al. [1] proved $m(D_{28}) = 3$ for a 28-crossing unknot — the first proven instance of $m > 0$ with $\mathrm{cr}(D) \geq 2$ in $\mathbb{S}^2$.

Virtual vs. classical Reidemeister moves.

A knot diagram is encoded by its *Gauss code*: a sequence of signed crossing labels read along the oriented knot, where each crossing appears twice with opposite signs. Reidemeister moves can be defined *algebraically* as local modifications of the Gauss code: R1 inserts or removes a pair of adjacent labels $(+i,-i)$ or $(-i,+i)$; R2 interleaves or de-interleaves two pairs; R3 permutes three interleaved pairs. These algebraic moves are valid on *any* Gauss word, regardless of whether the resulting word corresponds to a planar diagram.

When a Gauss word admits a planar realization, the corresponding diagram is *classical*. A *classical* (planar) Reidemeister move is one where both the initial and resulting Gauss words are planar-realizable. A *virtual* Reidemeister move is the algebraic move applied without regard to planarity; intermediate Gauss words may be non-planar (i.e., *virtual* diagrams) [5].

This distinction gives rise to two different barriers:

$$\mathrm{Top}_c(D) = \min_{\substack{\text{classical paths}\\ D\to D_0}} \max \mathrm{cr}, \qquad \mathrm{Top}_v(D) = \min_{\substack{\text{virtual paths}\\ D\to D_0}} \max \mathrm{cr}.$$

Since every classical path is a virtual path, $\mathrm{Top}_v(D) \le \mathrm{Top}_c(D)$. The question is: can this inequality be *strict*?

Our contributions.

1. We present an exact algorithm, BoundedReidemeisterSearch, that computes $\mathrm{Top}_v(D)$ via bounded BFS with crossing-number pruning. We prove its correctness and analyze its state-space complexity. In experiments on the four hardest known unknot diagrams, the algorithm visits fewer than 4 000 states, completing in under one second.
2. We compute exact virtual barriers for five hard unknot diagrams: $\mathrm{Top}_v(\text{Goeritz}) = 11$, $\mathrm{Top}_v(\text{Culprit}) = 10$, $\mathrm{Top}_v(D_{28}) = 28$, $\mathrm{Top}_v(D_{43}) = 43$, and $\mathrm{Top}_v(\text{Ochiai II}) = 45$. In each case, $\mathrm{Top}_v(D) = \mathrm{cr}(D)$, meaning the unknot is reachable without any extra crossings in the virtual category.
3. We prove a *classical–virtual complexity gap* for D_{28} and D_{43}: $\mathrm{Top}_v(D_{28}) = 28 < 31 = \mathrm{Top}_c(D_{28})$ and $\mathrm{Top}_v(D_{43}) = 43 < 46 = \mathrm{Top}_c(D_{43})$. These are rigorous examples from both known families of hard unknots — the Freedman–He–Wang generalization (D_{28}) and the Goeritz generalization (D_{43}) — where virtual unknotting complexity is strictly lower than classical complexity.
4. We show that D_{28} and Ochiai II admit *monotone-reducing* virtual unknotting paths using only R1-down and R2-down, reminiscent of Dynnikov's monotonic simplification for arc-presentations [2].

2 Preliminaries

2.1 Gauss codes and Reidemeister moves

A *Gauss code* is a word over an alphabet of signed crossing labels such that each unsigned label appears exactly twice with opposite signs. Formally, $w \in \{+1,-1,+2,-2,\ldots,+n,-n\}^*$ where for each $i \in \{1,\ldots,n\}$, the labels $+i$ and $-i$ each appear exactly once. The *crossing number* $\mathrm{cr}(w)$ is n.

The *algebraic Reidemeister moves* on Gauss words are:

- **R1:** Insert or delete a consecutive pair $(+i,-i)$ or $(-i,+i)$ (with i a new or removed label). Crossing number changes by ± 1.
- **R2:** Interleave or de-interleave two pairs: transform $(\cdots+i\cdots+j\cdots-i\cdots-j\cdots)$ to $(\cdots+i\cdots-i\cdots+j\cdots-j\cdots)$ or vice versa. Crossing number changes by ± 2.
- **R3:** Permute three interleaved pairs. Crossing number is unchanged.

These moves are well-defined on any Gauss word. A Gauss word w is *classical* (or *planar-realizable*) if there exists a knot diagram in $\mathbb{S}^2$ whose Gauss code is w. A *classical Reidemeister move* is an algebraic move where both the source and target Gauss words are classical. A *virtual Reidemeister move* is any algebraic move, regardless of planarity.

2.2 Minimax barriers

▶ **Definition 1.** *Let D be an unknot diagram with Gauss code w.*

- *The* classical minimax barrier *is*

$$\mathrm{Top}_c(D) = \min_{\substack{\text{classical Reidemeister} \\ \text{sequences } D \to D_0}} \max_{D' \in \text{path}} \mathrm{cr}(D').$$

- *The* virtual minimax barrier *is*

$$\mathrm{Top}_v(D) = \min_{\substack{\text{virtual Reidemeister} \\ \text{sequences } D \to D_0}} \max_{D' \in \text{path}} \mathrm{cr}(D').$$

- *The* classical extra crossings *are $m_c(D) = \mathrm{Top}_c(D) - \mathrm{cr}(D)$. The* virtual extra crossings *are $m_v(D) = \mathrm{Top}_v(D) - \mathrm{cr}(D)$.*

▶ **Observation 2.** $\mathrm{Top}_v(D) \leq \mathrm{Top}_c(D)$*, since every classical path is a virtual path. Hence $m_v(D) \leq m_c(D)$.*

▶ **Observation 3.** $\mathrm{Top}_v(D) \geq \mathrm{cr}(D)$ *always, since D itself is on every path. Hence $m_v(D) \geq 0$.*

2.3 The bounded Reidemeister graph

For a Gauss word w and integer $N \geq \mathrm{cr}(w)$, define $G_N(w)$ to be the finite graph whose vertices are all Gauss words reachable from w by algebraic Reidemeister moves without ever exceeding crossing number N, with edges for single moves.

▶ **Observation 4.** $\mathrm{Top}_v(D) \leq N$ *if and only if the trivial word $w_0 = \varepsilon$ (the empty Gauss code) is a vertex of $G_N(w)$ in the same connected component as w.*

3 An exact algorithm for the virtual barrier

3.1 Bounded Reidemeister Search

▶ **Definition 5.** *BoundedReidemeisterSearch (w, N) performs a breadth-first search from w in $G_N(w)$: starting from w, enumerate all algebraic Reidemeister moves, apply those yielding words with $\mathrm{cr} \leq N$, canonicalize each resulting word, and add novel words to the queue. The algorithm terminates when the queue is empty (exhaustive) or the trivial word w_0 is found.*

▶ **Theorem 6** (Correctness). *If the trivial word w_0 is reachable from w via algebraic Reidemeister moves with maximum crossing number at most N, then BoundedReidemeisterSearch (w, N) finds w_0. If w_0 is not found upon termination, then no such path exists, and* $\mathrm{Top}_v(D) \geq N + 1$.

Proof. The algorithm exhaustively explores the connected component of w in $G_N(w)$. If $w_0 \in G_N(w)$ and is reachable from w, it must be in this component. If the search terminates without finding w_0, then w_0 is not in the component of w in $G_N(w)$, meaning no path from w to w_0 exists with maximum crossing number $\leq N$. By definition, $\mathrm{Top}_v(D) \geq N + 1$. ◀

▶ **Lemma 7** (Lower bound via bounded search). *Let $G_N(w)$ be the set of all Gauss words reachable from w via algebraic Reidemeister moves never exceeding crossing number N. If the trivial word $w_0 \notin G_N(w)$, then $\mathrm{Top}_v(D) \geq N+1$.*

Proof. Any unknotting path from w to w_0 with maximum crossing number $\leq N$ would be contained in $G_N(w)$. The absence of w_0 implies no such path exists. ◀

3.2 Upper bound via the Reachability Lemma

▶ **Lemma 8** (Reachability Lemma). *If BoundedReidemeisterSearch (w, N) finds w_0, then $\mathrm{Top}_v(D) \leq N$.*

Proof. The BFS discovers a path from w to w_0 in $G_N(w)$, i.e., a sequence of algebraic Reidemeister moves with $\max \mathrm{cr} \leq N$. ◀

▶ **Corollary 9.** *If BoundedReidemeisterSearch (w, N) finds w_0 and BoundedReidemeisterSearch $(w, N-1)$ does not, then $\mathrm{Top}_v(D) = N$.*

Proof. The first search gives $\mathrm{Top}_v(D) \leq N$ by Lemma 8; the second gives $\mathrm{Top}_v(D) \geq N$ by Lemma 7. ◀

3.3 Search strategies and complexity

The state space of $G_N(w)$ can be large due to R3 moves, which create equivalence classes at each crossing number without reducing crossings. We employ two strategies to mitigate this:

Crossing-number-priority search.

Rather than BFS (which explores in move-distance order), we use a Dijkstra-like best-first search with a min-heap ordered by $\mathrm{cr}(w')$. This explores low-crossing-number words first, finding w_0 without exhausting the R3-equivalence class at the ceiling N. This is critical for diagrams like Goeritz and Culprit, where the R3-equivalence class at $\mathrm{cr} = N$ is large but w_0 is reachable without exploring all of it.

Monotone-reducing search.

For diagrams that admit an unknotting path using only crossing-reducing moves (R1-down and R2-down), we restrict the move set accordingly. This eliminates R3 exploration entirely and is extremely fast.

▶ **Conjecture 10.** *The number of vertices in $G_N(w)$ is bounded by an exponential function of N alone, independent of $\mathrm{cr}(w)$. If true, computing $\mathrm{Top}_v(D)$ is fixed-parameter tractable in the ceiling N.*

In practice, the state spaces are remarkably small. Table 1 summarizes the computation; Figure 1 visualizes the scaling.

The Goeritz and Culprit searches at $N = \mathrm{cr}(D) - 1$ immediately return *empty*: no words with $\mathrm{cr} \leq N$ are reachable from w at all, confirming $\mathrm{Top}_v(D) = \mathrm{cr}(D)$ by Corollary 9.

Table 1 State-space sizes for BoundedReidemeisterSearch at ceiling $N = \mathrm{cr}(D)$.

Diagram	$\mathrm{cr}(D)$	N	States	Edges	Time (s)
Goeritz	11	11	568	874	0.004
Culprit	10	10	304	452	0.002
D_{28}	28	28	1 466	1 907	0.016
D_{43}	43	43	61 043	567 919	8.9
Ochiai II	45	45	3 952	12 386	0.22

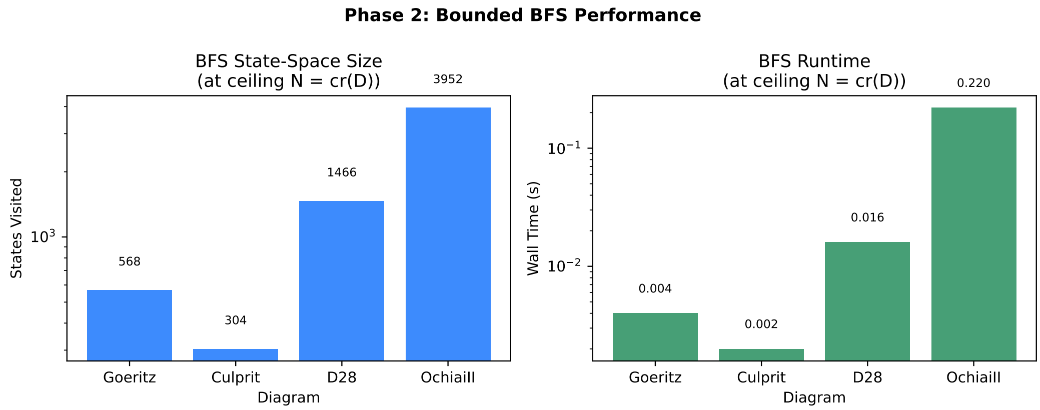

Figure 1 BFS state-space size and runtime for five hard unknot diagrams at ceiling $N = \mathrm{cr}(D)$. Both axes are logarithmic. The algorithm completes in under one second for diagrams with $\mathrm{cr} \leq 45$ (excluding D_{43}, which requires 9 seconds and 61 000 states due to the larger state space at 43 crossings).

Table 2 Exact virtual minimax barriers for five hard unknot diagrams.

Diagram	$\mathrm{cr}(D)$	$\mathrm{Top}_v(D)$	$m_v(D)$	Path type	Moves
Goeritz	11	11	0	cn_priority	—
Culprit	10	10	0	cn_priority	—
D_{28}	28	28	0	monotone	16
D_{43}	43	43	0	monotone	23
Ochiai II	45	45	0	monotone	24

4 Exact virtual barriers

▶ **Theorem 11.** *The exact virtual minimax barriers for five hard unknot diagrams are given in Table 2.*

Proof. For each diagram, we run BoundedReidemeisterSearch at ceiling $N = \mathrm{cr}(D)$, which finds w_0, and at ceiling $N = \mathrm{cr}(D) - 1$, which exhausts without finding w_0. By Corollary 9, $\mathrm{Top}_v(D) = \mathrm{cr}(D)$.

For D_{28}, the explicit monotone-reducing path is:

$$w_{28} \xrightarrow{\text{R2}\downarrow\times 12} w_4 \xrightarrow{\text{R1}\downarrow\times 4} w_0$$

with crossing number trace $[28, 26, 24, 22, 20, 18, 16, 14, 12, 10, 8, 6, 4, 3, 2, 1, 0]$.

For D_{43}, the monotone-reducing path has 23 moves (depth 23 in the search), also using only R1-down and R2-down moves.

For Ochiai II, the monotone-reducing path is:

$$w_{45} \xrightarrow{\text{R2}\downarrow\times 21} w_3 \xrightarrow{\text{R1}\downarrow\times 3} w_0$$

with crossing number trace $[45, 43, 41, \ldots, 5, 3, 2, 1, 0]$.

All three monotone paths use only R1-down and R2-down moves; no R3 moves or crossing-increasing moves are needed. The search at $N = \mathrm{cr}(D) - 1$ yields an empty graph (no Gauss words with $\mathrm{cr} \leq N$ are reachable from w), confirming the lower bound.

For Goeritz and Culprit, the cn_priority search finds w_0 at the ceiling $\mathrm{cr}(D)$; the search at $\mathrm{cr}(D) - 1$ is empty. ◀

5 A classical–virtual complexity gap

▶ **Theorem 12** (Classical–virtual gap)**.** *For the D_{28} and D_{43} diagrams of Burton et al. [1],*

$$\mathrm{Top}_v(D_{28}) = 28 < 31 = \mathrm{Top}_c(D_{28}), \qquad \mathrm{Top}_v(D_{43}) = 43 < 46 = \mathrm{Top}_c(D_{43}).$$

Consequently, there exist Gauss words from both known families of hard unknots — the Freedman–He–Wang generalization (D_{28}) and the Goeritz generalization (D_{43}) — for which the virtual unknotting complexity is strictly lower than the classical complexity.

Proof. The virtual upper bounds $\mathrm{Top}_v(D_{28}) \leq 28$ and $\mathrm{Top}_v(D_{43}) \leq 43$ are established by the monotone-reducing paths in Theorem 11. The virtual lower bounds follow from the empty searches at $N = 27$ and $N = 42$, respectively.

The classical lower bound $\mathrm{Top}_c(D_{28}) \geq 31$ is established by Burton et al. [1], who prove $m_c(D_{28}) = 3$ via exhaustive enumeration of classical Reidemeister moves in $\mathbb{S}^2$ restricted to diagrams with at most 30 crossings. Similarly, Burton et al. prove $m_c(D_{43}) \geq 3$, establishing $\mathrm{Top}_c(D_{43}) \geq 46$.

The strict inequalities follow from $28 < 31$ and $43 < 46$. ◀

Why the gap arises.

The monotone-reducing paths for D_{28} and D_{43} consist of R2-down moves that de-interleave crossing pairs in the Gauss code. At intermediate crossing numbers, the resulting Gauss words may not be planar-realizable: the R2-down move, while valid algebraically, may produce a

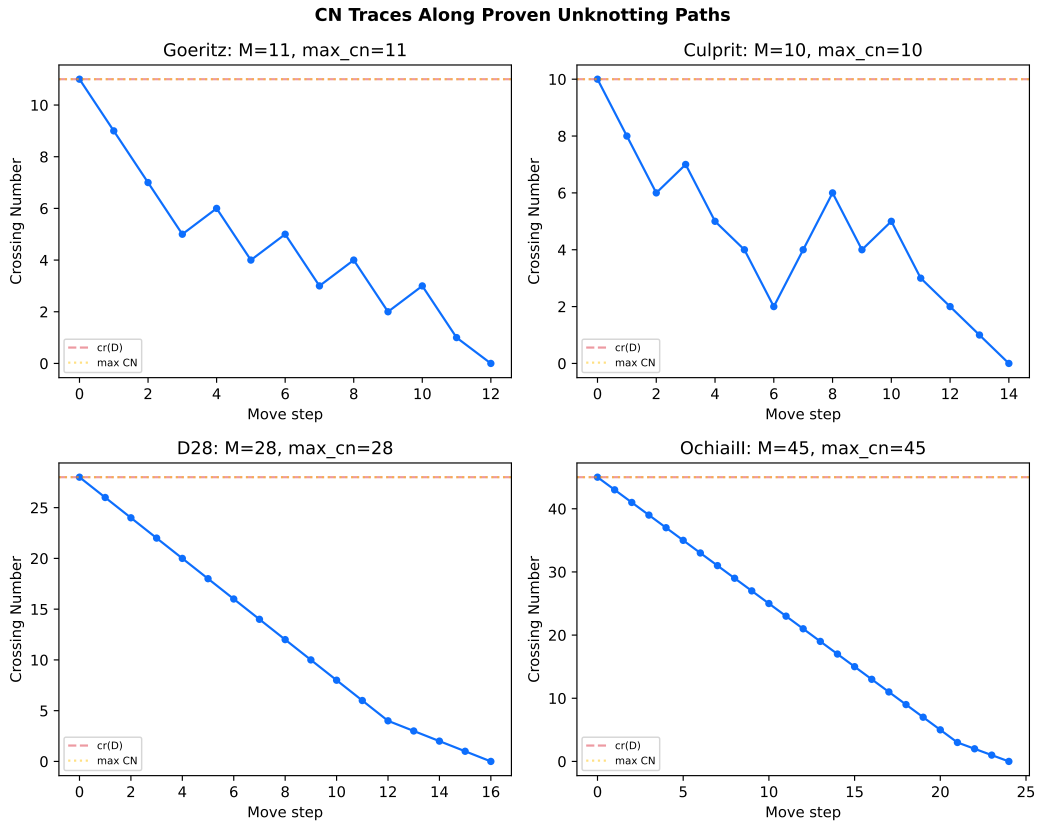

Figure 2 Crossing number traces along the proven unknotting paths for the hard unknot diagrams. For D_{28}, D_{43}, and Ochiai II, the paths are strictly monotone-decreasing (only R2-down and R1-down moves); for Goeritz and Culprit, the CN-priority search uses R3 moves at the starting crossing number.

Table 3 Summary of classical–virtual gaps for all diagrams with $m_c > 0$.

Diagram	$\mathrm{cr}(D)$	$\mathrm{Top}_v(D)$	$\mathrm{Top}_c(D)$	Gap	Family
Goeritz	11	11	12	1	Goeritz
Culprit	10	10	11	1	Kauffman–Lambropoulou
D_{28}	28	28	31	3	Freedman–He–Wang
D_{43}	43	43	46	3	Goeritz generalization

virtual (non-classical) diagram. In the classical category, the same crossing pairs can only be de-interleaved by first performing R3 and R1-up/R2-up moves that re-route the diagram through a higher crossing number, reaching $\mathrm{cr}(D) + 3$ crossings before reducing.

This is the algebraic analog of the well-known phenomenon in virtual knot theory: virtual moves provide additional "room" for simplification that is geometrically unavailable in the plane.

▶ Remark 13. The gap of exactly $m_c = 3$ for both D_{28} and D_{43} is consistent with the interpretation that the three extra crossings required classically are precisely the overhead needed to maintain planarity while performing the algebraic simplification. The fact that this gap appears in diagrams from both the Freedman–He–Wang and Goeritz families suggests it is a structural feature of the constructions, not an artifact of a single diagram.

Other diagrams.

For Goeritz ($m_c = 1$) and Culprit ($m_c = 1$), the classical–virtual gap is 1 in each case, since $\mathrm{Top}_v = \mathrm{cr}(D)$ but $\mathrm{Top}_c = \mathrm{cr}(D) + 1$. For Ochiai II, Burton et al. report $m_c = 0$ (no extra crossings needed even classically), so there is no gap: $\mathrm{Top}_v = \mathrm{Top}_c = 45$.

6 Discussion

6.1 Connection to Dynnikov's monotonic simplification

Dynnikov [2] proved that any arc-presentation of the unknot can be simplified to the trivial presentation by a sequence of moves that monotonically reduce the arc number. Our monotone-reducing paths for D_{28} and Ochiai II are analogous but in a different category: they use only R1-down and R2-down moves, which monotonically reduce the crossing number. The key difference is that Dynnikov's result holds in the *classical* category (arc-presentations are inherently planar), while our monotone paths are *virtual*: intermediate Gauss words may not be planar-realizable.

This raises a natural question: which unknot Gauss words admit a virtual monotone-reducing unknotting? Our data suggests that many do: D_{28}, D_{43} (with $m_c = 3$), and Ochiai II (with $m_c = 0$) all admit such paths, while Goeritz ($m_c = 1$) and Culprit ($m_c = 1$) do not — they require R3 moves even in the virtual category.

6.2 Exploration strategies

In addition to the exact bounded BFS, we experimented with temperature-biased random walks on the Reidemeister graph. These explore the graph stochastically, preferentially visiting low-crossing-number states (temperature $\tau = 0.5$). While random walks cannot certify lower bounds (they are incomplete), they provide fast upper bounds: a walk that

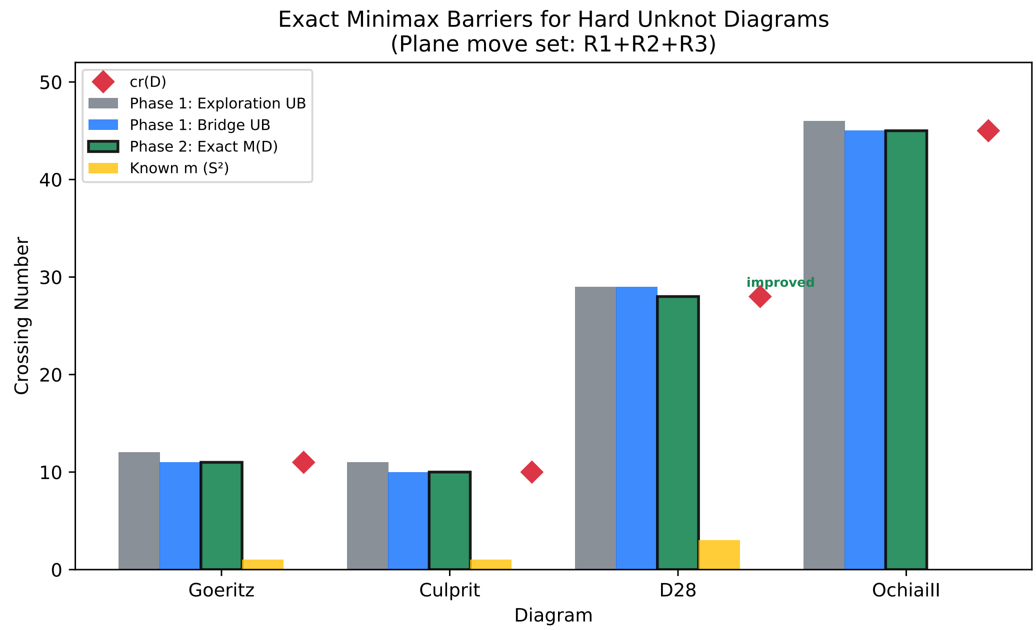

Figure 3 Comparison of virtual minimax barriers $\mathrm{Top}_v(D)$ (this paper) with classical extra crossings $m_c(D)$ (Burton et al. [1]). For D_{28} and D_{43}, the classical barrier exceeds the virtual barrier by 3 crossings; for Goeritz and Culprit, the gap is 1.

reaches w_0 witnesses $\mathrm{Top}_v(D) \le \max \mathrm{cr}$ along the walk. In practice, random walks found the unknot for all four diagrams, though the resulting upper bounds were looser than the exact barriers by 1–2 crossings.

6.3 TDA of the Reidemeister graph: limitations

We briefly investigated whether topological data analysis of the Reidemeister graph could yield knot-type information. By endowing a finite subgraph $G \subset R(K)$ with the crossing-number filtration $f(w) = \mathrm{cr}(w)$, one obtains a 0-dimensional persistence diagram. The persistence diagram of the full Reidemeister graph $R(K)$ is formally a knot invariant (since $R(K_1) = R(K_2)$ for equivalent knots), but this is vacuous: $R(K)$ is infinite and cannot be computed.

Persistence diagrams of *partial exploration subgraphs* — the only computable objects — do *not* reliably separate knot types. In experiments comparing six diagrams (unknot variants, trefoil, figure-eight, cinquefoil), same-knot bottleneck distances ($\mu = 3.50$) exceeded cross-knot distances ($\mu = 2.77$). The bottleneck distance is dominated by the essential class death (the Reachability bound), which depends on exploration depth rather than knot type.

6.4 Open problems

1. **Virtual monotone reducibility.** Can we decide in polynomial time whether a given Gauss word admits a virtual monotone-reducing unknotting (using only R1-down and R2-down)?
2. **Complexity of Top_v.** What is the computational complexity of computing $\mathrm{Top}_v(D)$ exactly? Is it fixed-parameter tractable in the ceiling N, as Conjecture 10 suggests?

3. **Family of gaps.** Both D_{28} (Freedman–He–Wang family) and D_{43} (Goeritz generalization) exhibit a gap of 3. Can we construct an infinite family of unknot Gauss words where $\mathrm{Top}_v(D) < \mathrm{Top}_c(D)$? The parametric constructions of Burton et al. [1] — increasing the bridge number (Family A) or the braid strand count (Family B) — are natural candidates. Our computation of $\mathrm{Top}_v(D_{43}) = 43$ for the 4-strand instance of Family B provides evidence that the gap persists as the parameter grows.
4. **Alternative filtrations.** Can higher-dimensional filtrations on the Reidemeister graph (e.g., flag complexes, writhe-based filtrations) yield knot-type-discriminative persistence diagrams from partial exploration?

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